PV-Finder: additional studies requested for ANA-IDTR-2026-01

Status on 10 November 2026. HL-LHC numbers: 1 920 PU200 $t\bar{t}$ events (r16443, $185 \leq \mu \leq 200$). Errors: paired event bootstrap or Clopper–Pearson.

Study	Note	Status	Result / next step
Matching window	2.4, 4.1	Done	Keep 0.1694 mm ($3\times$ the 68% position residual) for all vertices, 0.1 mm for HS only. Code default is now 0.1694 mm. Audit lists every other window used in the note; the GNN finder-cap and oracle bars still need a rerun at 0.1694 mm.
Split vertices in efficiency	2.4.2, 4.1	Done	Cannot change the efficiency: with closest-first matching, every truth vertex next to a split vertex is already matched (proven by tests). PV-Finder 83.16% vs AMVF 79.80%, $+3.36 \pm 0.06$ points; at 0.1 mm $+3.12 \pm 0.06$.
HS studies vs pile-up	4.1, 7.4–7.5	Done	HS reco (0.1 mm): 99.9% / 99.8%. Clean HS: GNN 96.4%, AMVF 92.5%. HS tracks on HS vertex: 74.9% / 74.1% (purity 91.3% / 88.4%). Reco+selected: GNN 96.3%, AMVF 92.9%. All vs μ in ATLAS style.
AMVF HS selection vs ITk ∼93%	7.5	Done	Reconciled: the stored AMVF primary gives $92.9 \pm 0.6\%$. The quoted 80.7% re-ranked the stored track lists by $\sum p_{\mathrm{T}}^2$, which picks a different vertex in 428/1 920 events.
ITk HS longitudinal resolution	4.1.1	Done	Athena iterative-RMS definition: PV-Finder 12.4 ± 0.4 µm, AMVF 10.2 ± 0.3 µm (reproduces ITk ≈ 10 µm), flat in μ. PV-Finder is probably limited by its 40 µm output bins.
Run 2 vs Run 4 gap	6.1	Partial	Not a methodology effect (binning, range, statistics, events: < 0.06 in the ratio). AMVF/PV-Finder $\sigma_{\text{vtx-vtx}}$: 1.93 (Run 2), 1.33 (PU200). PV-Finder scales with detector precision; AMVF improved more, consistent with an extra limit in its Run 2 configuration (not verifiable from the ntuples).
Data–MC validation	6.2, 7.6	Partial	No Run 3 MC exists (only Run 2 $t\bar{t}$). MC now on all Fig. 6.2 panels. AMVF as reference in MC biases recovery by $+2.4 \pm 0.1$ points and explains the data–MC gap in unmatched peaks. Run 3 needs mc23 $t\bar{t}$ ntuples.
z_0 association baseline	7.1	Done	Added to Fig. 7.2 (clean/truth): nearest z_0 31.9%, within $3\sigma(z_0)$ 33.1%, AMVF 30.9%, GNN 68.4%. Caption drafted.
Fig. 7.2 and HS vs pile-up	7.5	Done	Stages and HS metrics in 10 μ bins. The full-μ GNN keeps HS selection at 94–98% in every bin; the PU200-trained GNN drops at low μ.
Timing vs pile-up	5.1.2	Blocked	Needs per-event Athena timing (algorithm, device, μ, ms) from the integration team. Plotting code ready.
AMVF weights vs GNN	7.4	Declined	Weights are not stored in our ntuples, and a comparison in AMVF-weight space would favour AMVF. Paragraph drafted explaining this.
GNN on more data	7.3	Queued	Starts after the small-model study.
Paper figures, Table 1	–	Partial	Figs. 1–4 and Table 1 (= ITk paper Table 6) done, captions corrected. Figs. 5–6 need Run 3 MC; Fig. 7 needs timing data.
Small model for CPU	5	Running	Exact packed-track inference: $1.4\times$ faster for free. CPU speed-up vs production: s5 $6.4\times$, s8 $9.0\times$. s5 (1+1 epochs) held-out: efficiency -0.33 ± 0.04 / -0.26 ± 0.04 points, -0.8 fakes/event, σ $+5$ µm: inside the 0.5 limit, misses the 0.2 target. s5 on the full 2+2 schedule running (∼16 h); s8 still training. INT8 gives $\leq 1.25\times$.
Repositories	–	Partial	GitHub up to date. GitLab note unreachable from sneezy; note edits drafted, to apply via lxplus.

Needs input: Run 3 mc23 $t\bar{t}$ ntuples with truth and AMVF vertices; per-event Athena timing; confirmation of the “Run 2 MC” campaign label; GitLab access for the note edits.